



Architecture, Infrastructure, Security and Data Privacy BrokerVerse OOTB

Prepared by – **iorta TechNXT**

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1. Document Control

Date	Version	Author & Business Unit Info	Reviewed By & Business Unit Info	Approved By & Business Unit Info	Change History
03 October 2026	1.0	iorta TechNXT			Initial issue

2. Acronyms

Acronym / Symbol	Definition / Meaning
OOTB	Out of the box
IC	Insurance Commission
NPC	National Privacy Commission
DPA	Data Privacy Act of 2012 (Republic Act 10173)
IRR	Implementing Rules and Regulations
DPO	Data Protection Officer
PIC	Personal information controller
PIP	Personal information processor
SPA	Single-page application
API	Application programming interface
RDS	Amazon Relational Database Service
EFS	Amazon Elastic File System
WAF	Web application firewall
CDN	Content delivery network
RTO	Recovery time objective
RPO	Recovery point objective
PITR	Point-in-time recovery
TOTP	Time-based one-time password
JWT	JSON Web Token

SoD	Segregation of duties
BIR	Bureau of Internal Revenue
DR	Disaster recovery
UAT	User acceptance testing

3. Introduction

3.1 Purpose

This document describes how BrokerVerse OOTB is built, how it can be hosted for a Philippine non-life insurance broker, how it is secured and how it supports the broker's obligations under the Data Privacy Act of 2012. It is written for the broker's management, IT, compliance and data protection officer, and for the support team that takes over the system.

3.2 Sources and conventions

The facts in this document come from the source code and configuration of branch `brokerverse-platform` (backend in `backend/`, front end in `brokerverse/`), the deployment package in `deploy/`, the GitHub workflows in `.github/workflows/`, `docker-compose.yml`, the onboarding guides in `docs/onboarding/` and the eleven Solution Architecture documents in `docs/architecture/` (version 1.1, 29 September 2026).

- Statements without a label describe what the system does today.
- **Recommended** marks a design choice, a size, a threshold or a procedure that iorta TechNXT advises but that is not built into the code or the deployment package. It must be confirmed by the broker and the hosting team.
- **Indicative** marks a cost or size estimate. Indicative figures are planning ranges, not quotations.

- **Gap** marks a point where the current build or pipeline falls short of good practice and needs action before or after go-live.

This document gives the regulatory context in general terms. It is not legal advice; the broker's compliance officer and DPO confirm how each rule applies to the broker.

3.3 Related documents

Document	Content
01 Solution Component Diagram and Application Architecture	Components, request pipeline, modules, integrations
02 and 03 Database Design and Inventory	Tables, keys, retention classes
04 Technology Stack	Versions, licences, lifecycle notes
05 Shared Service Components	Authentication, settings, audit, numbering, PDF engine
06 to 11	Capacity, high availability, RTO and RPO, backup, archival, monitoring
deploy/README.md, deploy/REFERENCE.md	Deployment checklist, environment variables, start-up rules
docs/onboarding/	Getting started, go-live data set-up, UAT scripts, support and escalation

4. Solution overview and logical architecture

4.1 What BrokerVerse does

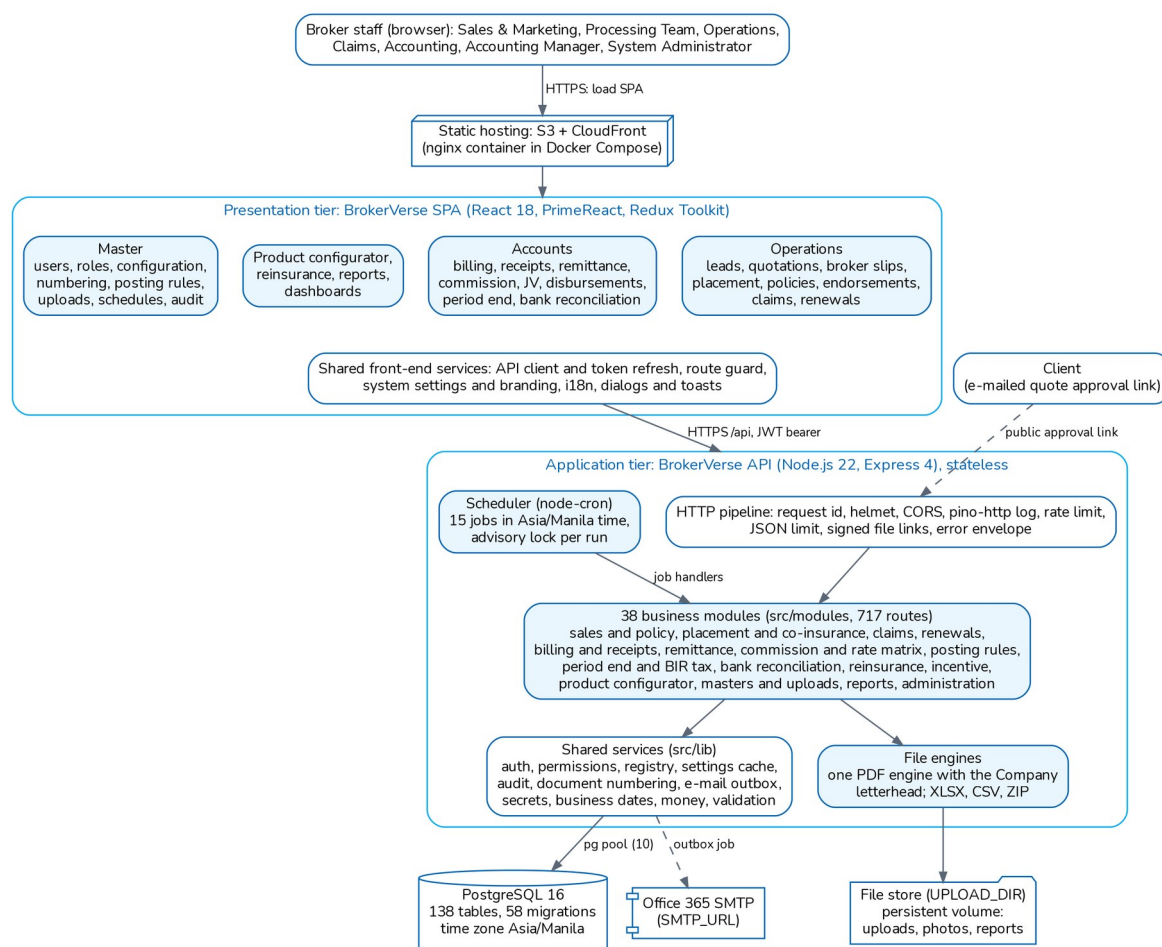
BrokerVerse OOTB is the operating system of a non-life insurance broker. It covers the full broking cycle: leads, quotations and quick quotes, broker slips to several insurers, placement and co-insurance, policy issue, endorsements, claims, renewals, billing, receipts, collections and credit control, remittance to insurers, direct bill, commission and incentives, reinsurance, bank and insurer statement reconciliation, period-end close with BIR tax outputs, reports and dashboards. Referrers and sub-agents do not sign in; their business is entered by Sales & Marketing.

4.2 Architecture at a glance

Aspect	BrokerVerse OOTB
Style	Three-tier web application: SPA, stateless REST API, relational database. The API is a modular monolith.
Presentation tier	React 18 SPA (PrimeReact, Redux Toolkit), served as static files from object storage and a CDN, or from an nginx container
Application tier	Node.js 22 with Express 4. 45 module folders under /api, 868 registered routes, two health endpoints, a built-in scheduler of 18 jobs in Asia/Manila time
Data tier	PostgreSQL 16, database time zone Asia/Manila, 80 migration files applied on start under an advisory lock
File store	Persistent volume at UPLOAD_DIR (documents, ID images, photos, generated reports)
Security	JWT sessions with refresh-token rotation, role and permission checks on every route, maker-checker, authority matrix, SoD rules, access reviews, TOTP two-factor, audit log

Route, module and migration counts were taken from the code on 03 October 2026. The diagrams reproduced from the architecture set show the counts of 29 September 2026 (38 modules, 717 routes, 58 migrations); the structure is unchanged.

4.3 Logical components



Logical component view of BrokerVerse OOTB (architecture document 01)

Layer	Components
Users	Broker staff in a browser; clients through public links (quotation approval, payment checkout)
Edge	DNS, TLS, CDN or reverse proxy, WAF (recommended)
Presentation	SPA: Operations, Accounts, Master, Product Configurator, Reinsurance, Reports, Dashboards
HTTP pipeline	Request id, security headers (helmet), CORS, access log with redaction, rate limit, JSON size limit, signed file links, one error envelope
Business modules	backend/src/modules: each module owns its routes, services and tables
Shared services	backend/src/lib: authentication, permissions, settings cache, audit, numbering, e-mail outbox, secrets, PDF engine, letterhead
Scheduler	node-cron jobs with a PostgreSQL advisory lock per run, so each run executes once across instances
Data	PostgreSQL 16 and the file store

4.4 Business modules

Area	Backend modules
Sales and operations	leads, clients, addresses, quotations, packages, premium-charges, placement, policies, endorsements, documents
Claims and renewals	claims, renewals
Billing and collection	receipts, collections, credit-control, payments, payment-

Area	Backend modules
	gateway
Insurer accounts	remittance, insurer-reconciliation, commission, commission-rates, incentive, reinsurance
Accounting	accounting, posting-rules, journal-vouchers, disbursements, period-end, bank-reconciliation
Configuration	product-configurator, masters, document-numbering, settings, system-settings, schedules
Administration and security	auth, users, access-control, notifications, uploads, search, dashboard, reports, system

4.5 Integrations

Integration	Direction	How it works
E-mail (SMTP)	Outbound	Messages are queued in email_outbox; the email-outbox job sends up to 50 per run every 5 minutes through SMTP_URL (Office 365, port 587, STARTTLS), up to 5 attempts. Off until "Send e-mails" is switched on in Master > Configuration.
Payment gateways	Inbound and outbound	Payment links (PL- numbers) for a quotation, a package quotation or a policy's open premium. Providers: SANDBOX (training and UAT), PayMongo (Checkout Session) and Dragonpay (Payment Switch). Webhooks are signature-checked, logged in payment_events and idempotent; a paid link creates the official receipt and can issue a package policy.
Insurers	Outbound files and e-mail	Broker slips, placement orders, remittance statements, commission debit notes as PDF, CSV or XLSX. No insurer API.
Insurer statements	Inbound files	CSV or XLSX statements of account read with an insurer statement format (column mapping per insurer) for Accounts > Insurer Reconciliation.
Bank statements	Inbound files	CSV or XLSX statements read with a bank statement format; BDO, BPI, Metrobank and a generic layout ship as standard.
Electronic transfers	Recorded	Transfers to insurers are approved in the system and the bank result is recorded (Completed or Failed). No bank payment file or bank API is generated.
Reinsurers	Outbound files	Bordereaux and reconciliation files as CSV
Uploads and go-live imports	Inbound files	10 MB per file, 50 MB decompressed, 20,000 rows; templates in docs/templates
Document generation	Internal	One PDF engine (lib/pdf) with the letterhead of the primary company in the Company master; XLSX, CSV and ZIP writers. No third-party document library.
Client quotation approval	Inbound (public)	Signed link valid for quotations.approval_link_ttl_hours (168 hours)

Recommended: Before PayMongo or Dragonpay are switched to live, obtain the merchant accounts and test against the providers' own sandboxes. The automated tests cover the SANDBOX provider and signed test payloads only. Gateway fees are recorded on the link but not journalised.

5. Technology stack and versions

Versions are the exact versions resolved in backend/package-lock.json and brokerverse/package-lock.json on 03 October 2026.

5.1 Runtime and platform

Component	Version	Use
Node.js	22 (node:22-alpine; engines >=22)	API runtime and build
PostgreSQL	16 (postgres:16-alpine in Compose, postgres:16 in CI)	System of record, extension pgcrypto
nginx	1.27-alpine	Web container (Compose); reverse proxy on the EC2 host
PM2	Latest from npm at deployment	Process manager on the EC2 deployment (deploy/ec2/ecosystem.config.cjs)
Docker and Docker Compose	Compose file without a version key	Single-server, test and trial installations
GitHub Actions	checkout, setup-node, upload and download-artifact, configure-aws-credentials v4; appleboy/ssh-action 1.2.0	CI and deployment

5.2 Backend dependencies

Package	Version	Purpose
express	4.22.3	HTTP server and routing
pg	8.23.0	PostgreSQL client, pool of 10 connections
zod	3.25.76	Request validation
pino / pino-http	9.14.0 / 10.5.0	JSON logs with redaction
helmet	8.3.0	Security headers
cors	2.8.6	Cross-origin policy
jsonwebtoken	9.0.3	Access, refresh and link tokens (HS256)
bcryptjs	2.4.3	Password hashing
multer	1.4.5-lts.2	Multipart uploads in memory with limits
nodemailer	10.0.12	SMTP delivery
node-cron	4.6.0	Job schedules
dotenv	16.6.1	.env loading
vitest / supertest	2.1.9 / 7.3.0	Integration tests against PostgreSQL
eslint	9.39.5	Lint

5.3 Front-end dependencies

Package	Version	Purpose
react, react-dom	18.2.0	UI library
react-scripts / @craco/craco	5.0.1 / 7.1.0	Build tool chain
primereact, primeflex, primeicons	10.3.1, 3.3.1, 6.0.1	Components, layout, icons
@reduxjs/toolkit, react-redux	2.0.1, 9.0.4	Application state
react-router-dom	6.30.6	Routing
axios	1.20.0	HTTP client with token refresh
formik	2.4.5	Forms
i18next, react-i18next	25.8.13, 16.5.4	English (complete), Thai (partial)
chart.js, @fullcalendar	4.5.1, 6.1.x	Charts and calendars
@fontsource/nunito	5.3.0	Bundled font, no external font service

The backend has 13 direct runtime dependencies. PDF, XLSX, ZIP, CSV (with formula-injection guarding), TOTP, AES-256-GCM encryption, HMAC link signing and rate limiting are part of the code base. All direct dependencies use permissive licences (MIT, MIT-0, ISC, BSD-2-Clause, Apache-2.0, OFL-1.1).

6. Deployment architecture options

6.1 Common building blocks

Whatever the hosting option, BrokerVerse needs the same components.

Component	Requirement
Web hosting	Static files of the SPA (<code>npm run build</code>), with a fallback of every path to <code>index.html</code> ; <code>REACT_APP_BASE_URL</code> compiled in
API runtime	Container from backend/Dockerfile (runs as user node, port 8000, health check <code>/api/health</code>) or Node.js 22 under a process manager
Database	PostgreSQL 16, empty database and a login that may create tables; time zone Asia/Manila; TLS on the connection
File store	Persistent volume at <code>UPLOAD_DIR</code> , shared between instances when there is more than one
Secrets	<code>JWT_SECRET</code> , <code>DATA_ENCRYPTION_KEY</code> , <code>ADMIN_PASSWORD</code> , <code>DATABASE_URL</code> , <code>SMTP_URL</code> , payment gateway keys, in a secret store
TLS	HTTPS at the CDN, load balancer or reverse proxy
Outbound	SMTP (port 587) and, when used, the payment gateways over HTTPS
Inbound public paths	Quotation approval, payment checkout and gateway webhooks under <code>/api/public/...</code>

6.2 Environment layout

Environment	Purpose	Data	Recommended form
DEV	Development and integration by iorta TechNXT	Sample data (<code>SEED_SAMPLE_DATA=true</code>)	Single server or Docker Compose
UAT	Business acceptance, training, performance test	Synthetic UAT data (<code>backend/scripts/uat-scenario.js</code>) or masked copies	Same topology as PROD at smaller size
PROD	Live operations	Reference data only, then go-live imports	Highly available (two API instances, managed PostgreSQL with standby)
DR	Recovery after a regional or site disaster	Restored from cross-region or off-site backups	Pilot light or warm standby (chapter on resilience)

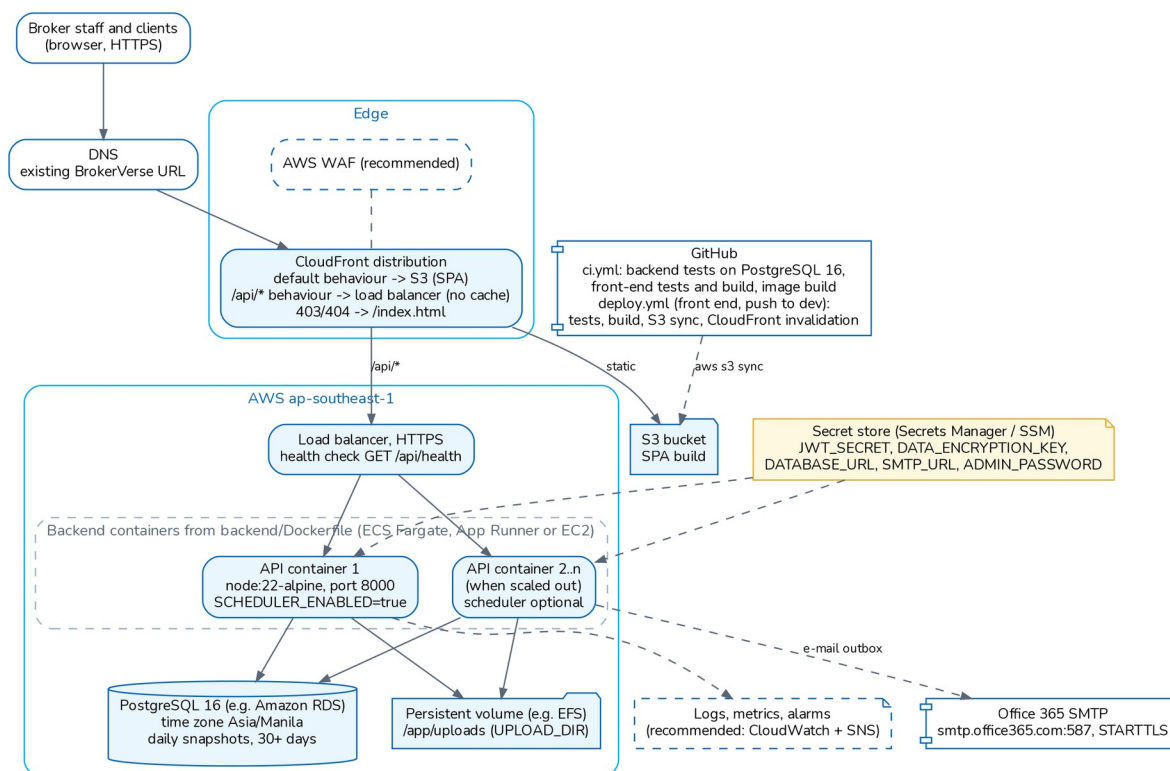
Recommended: Never copy production personal data into DEV or UAT without masking. Each environment has its own secrets; `DATA_ENCRYPTION_KEY` and `JWT_SECRET` must differ between environments.

6.3 Network zones

Zone	Contents	Allowed traffic
Public edge	DNS, CDN, WAF, public load balancer or reverse proxy	HTTPS 443 from the internet
Application zone (private)	API instances	From the load balancer only, on the API port; outbound to SMTP and the payment gateways
Data zone (private)	PostgreSQL, file store	PostgreSQL 5432 and NFS from the application zone only
Management zone	Bastion or session manager, CI/CD runner access	Administrator access with multi-factor sign-in, logged

6.4 Option A: customer cloud on AWS (Singapore)

This is the reference deployment of `deploy/README.md` and architecture documents 01 and 07. The region is `ap-southeast-1` (Singapore).



Production deployment on AWS ap-southeast-1 (architecture document 01)

Component	AWS service
Web hosting and CDN	Amazon S3 bucket and CloudFront; custom error responses 403 and 404 return <code>/index.html</code>
API address	Same-domain option: CloudFront behaviour <code>/api/*</code> to the load balancer, no caching
API	ECS Fargate (recommended), App Runner or EC2, behind an Application Load Balancer with HTTPS (ACM certificate)
Database	Amazon RDS for PostgreSQL 16, Multi-AZ, encrypted with KMS, <code>sslmode=require</code>
File store	Amazon EFS mounted at <code>/app/uploads</code> , encrypted at rest
Backups	RDS automated backups (35 days, PITR), AWS Backup plans for RDS and EFS, cross-region copies
Secrets	AWS Secrets Manager or SSM Parameter Store
Monitoring	CloudWatch Logs, metrics, alarms, Synthetics; SNS for notifications
WAF	AWS WAF on CloudFront with managed rule groups and rate-based rules
Image registry	Amazon ECR, replicated to the recovery region

Philippine data residency. The Data Privacy Act does not require personal data to be stored in the Philippines; it makes the broker accountable for personal data transferred abroad (DPA section 21). Hosting in Singapore is therefore a cross-border transfer that the broker documents and covers by contract (chapter on data privacy).

AWS has a Local Zone in Manila with a limited set of services (compute and block storage); managed RDS, EFS and the other services above run in the Singapore parent region. A broker that must keep the system of record in the Philippines uses option C, or confirms with AWS which services are available in Manila at the time of the decision.

6.4.1 Current demo deployment

The repository also deploys a demo system: `.github/workflows/deploy-frontend.yml` publishes the front end to S3 and CloudFront and reloads the API on one EC2 host with PM2 (`deploy/ec2/deploy.sh`). The API runs as one PM2 process (fork mode, restart above 700 MB) on port 8001 behind an nginx site that only forwards `/api/`. This is a single-instance set-up with no redundancy; it is suitable for demonstration and UAT, not as the production form of option A.

6.5 Option B: Microsoft Azure (Southeast Asia)

The Azure region Southeast Asia is in Singapore, so the data residency position is the same as for option A. The application needs no code change on Azure: it uses a container, PostgreSQL and a mounted file system.

Component	Azure service (recommended mapping)
Web hosting and CDN	Azure Storage static website with Azure Front Door (CDN and routing of <code>/api/*</code>)
WAF	Azure Front Door WAF policy
API	Azure Container Apps or App Service for Containers, two or more replicas across availability zones
Database	Azure Database for PostgreSQL Flexible Server 16, zone-redundant high availability
File store	Azure Files (SMB or NFS share) mounted at <code>/app/uploads</code>
Backups	Flexible Server automated backups with PITR (up to 35 days), geo-redundant backup storage; Azure Backup for Azure Files
Secrets	Azure Key Vault, referenced by the container app
Monitoring	Azure Monitor, Log Analytics, Application Insights availability tests
Registry	Azure Container Registry
Network	Virtual network with separate subnets for the application and the database (private access)

Recommended: The GitHub workflows in the repository target AWS. For Azure, the deployment steps (registry push, container app revision, static website upload, Front Door purge) must be added to the pipeline. Set the database time zone to Asia/Manila (server parameter `timezone`) or add the time zone option to `DATABASE_URL`, as done for Railway.

6.6 Option C: local partner or on-premise data centre in the Philippines

For a broker that wants the data held in the Philippines, BrokerVerse runs on virtual machines in a local data centre or with a Philippine hosting partner. The Docker Compose file and the EC2 scripts show both ways to run it without cloud managed services.

Component	On-premise or partner form
Web hosting	nginx serving the SPA build (the web image of Docker Compose), behind the TLS reverse proxy

Component	On-premise or partner form
Reverse proxy and WAF	Two nginx or HAProxy nodes with a virtual IP, TLS certificate; WAF appliance or ModSecurity with the OWASP Core Rule Set
API	Two VMs running the API container (or Node.js 22 with PM2)
Database	PostgreSQL 16 primary and streaming-replica standby on separate hosts; failover with Patroni or a manual runbook
File store	NFS share on redundant storage (or a clustered file system) mounted on both API VMs
Backups	Daily pg_dump -Fc and WAL archiving (pgBackRest or WAL-G) to a backup server; daily archive of the upload share; an off-site copy in a second Philippine site
Secrets	HashiCorp Vault, or an encrypted environment file readable only by the service account, with an escrow copy under dual control
Monitoring	Prometheus and Grafana or Zabbix, log shipping to a central log server or SIEM, external uptime check
DR	Second data centre in another Philippine city with replicated backups

Recommended: Ask the partner for evidence of physical security, power and cooling redundancy, and an ISO/IEC 27001 certificate or equivalent. Agree in the contract who patches the operating system, PostgreSQL and the reverse proxy.

6.7 Sizing by broker size

The sizes below extend the recommendation of architecture document 06 (200 named users, 80 concurrent, 50,000 policies a year: two API instances of 1 vCPU and 2 GB, database of 2 vCPU and 8 GB Multi-AZ). They apply to PROD and must be confirmed with the UAT performance test.

Recommended / Indicative: Assumptions per broker size: small up to 25 named users and about 5,000 policies a year; medium 26 to 100 users and about 20,000 policies; large 101 to 300 users and about 60,000 policies. Uploaded documents average 2 MB per policy. Replace these with the broker's own forecast.

Item	Small (up to 25 users)	Medium (26 to 100)	Large (101 to 300)
API instances	2 x 1 vCPU, 2 GB	2 x 1 vCPU, 2 GB	3 to 4 x 2 vCPU, 4 GB
Database	2 vCPU, 4 GB, standby	2 vCPU, 8 GB, standby	4 vCPU, 16 GB, standby
Database storage	50 GB	100 GB	250 GB
File store (year 1)	25 GB	60 GB	150 GB
Backup storage	100 GB	250 GB	600 GB
Concurrent users at peak	10	40	120

6.8 Indicative monthly cost

Indicative: Estimates of on-demand list prices for the PROD environment only, converted at an assumed PHP 58.00 to USD 1.00, before tax, support plans and data transfer above normal use. Reserved capacity or savings plans reduce them by roughly 25% to 40%. Obtain a quotation from the provider or partner before ordering.

Option	Small	Medium	Large
A: AWS Singapore	PHP 15,000 to 25,000	PHP 30,000 to 50,000	PHP 65,000 to 100,000
B: Azure Southeast Asia	PHP 15,000 to 27,000	PHP 30,000 to 55,000	PHP 65,000 to 105,000

Option	Small	Medium	Large
C: Philippine partner (managed VMs)	PHP 25,000 to 45,000	PHP 45,000 to 80,000	PHP 90,000 to 160,000

Additional environment	Indicative share of the PROD cost
UAT (single API instance, database without standby)	30% to 40%
DEV (single small server or Compose)	10% to 15%
DR as pilot light (backups and images in the second region or site, nothing running)	5% to 10%
DR as warm standby (small database replica and one API instance running)	30% to 50%

The largest cost items are the database with its standby, the load balancer or reverse proxy, the WAF and the log volume. The file store and the backups grow with the number of uploaded documents.

7. CI/CD and release management

7.1 Workflows in the repository

Workflow	Trigger	What it does
<code>.github/workflows/ci.yml</code> (CI)	Every pull request	Backend: <code>npm ci</code> , <code>npm run lint</code> , <code>npm test</code> against a <code>postgres:16</code> service. Front end: <code>npm ci --legacy-peer-deps</code> , <code>craco test</code> .
<code>.github/workflows/deploy-frontend.yml</code> (CI/CD Pipeline)	Push to <code>brokerverse-platform</code>	Builds the front end with Node 22 (<code>REACT_APP_BASE_URL</code> from the repository variable), publishes it to S3 (hashed files cached one year, <code>index.html</code> not cached), invalidates CloudFront, then reloads the API on the EC2 host over SSH (<code>deploy/ec2/deploy.sh</code>).
<code>brokerverse/.github/workflows/deploy.yml</code>	Push or pull request to <code>brokerverse-platform</code> , when the front end is its own repository	Front-end tests, check that <code>REACT_APP_BASE_URL</code> is set, build, check that <code>build/index.html</code> is a production build; the publish job runs only for a push to dev

The EC2 deployment script checks out the pushed branch, installs production dependencies with `npm ci --omit=dev`, creates the database if missing (`scripts/create-database.js`), installs the nginx site after `nginx -t`, starts the API under PM2 and waits up to two minutes for `/api/health` to return 200; otherwise it prints the last 80 log lines and fails the job.

Gap: In `deploy-frontend.yml` the lint and test jobs are commented out, so a push to `brokerverse-platform` deploys without running tests. Tests run only on pull requests (`ci.yml`). **Recommended:** protect the branch so that changes arrive only through pull requests with a green CI run, and restore the test jobs as a dependency of both deploy jobs.

Gap: The pipelines use long-lived AWS access keys and an SSH private key stored as GitHub secrets, and the backend image is not pushed to a registry. **Recommended:** GitHub OIDC with an IAM role limited to the bucket and the distribution; build, scan and push a versioned backend image (for example to ECR) and deploy that image, so a release can be rolled back by tag.

Note: In `brokerverse/.github/workflows/deploy.yml` the publish job requires the branch `dev` while the trigger is `brokerverse-platform`; as written it tests and builds but does not publish. Align the branch names when the front end is moved to its own repository.

7.2 Release process

Recommended release steps for UAT and PROD:

1. Merge to the release branch through a reviewed pull request with a green CI run.
2. Tag the release (for example `v1.0.0`, as named in `deploy/README.md`) and record `GIT_COMMIT` in the build so `GET /api/version` shows it.

3. Deploy to UAT, run the UAT scripts in docs/onboarding/UAT_SCRIPTS.md for the changed areas and obtain business sign-off.
4. Before PROD: take a manual database snapshot when the release contains migrations.
5. Deploy one API instance first; when /api/health is ready, roll the others.
6. Smoke test as in deploy/README.md section 4; keep the previous image and front-end build for rollback.

7.3 Environment configuration and secrets

Variable	Use	Rule enforced at production start
NODE_ENV	production turns on the start-up checks	
DATABASE_URL	PostgreSQL connection, sslmode=require on managed databases	
JWT_SECRET	Signs sessions, file links and approval tokens	Required, not a placeholder, at least 32 characters
DATA_ENCRYPTION_KEY	Encrypts TOTP secrets, keys reset-code hashes	Required, at least 32 characters, different from JWT_SECRET
ADMIN_PASSWORD	First password of the BrokerVerse administrator	Must meet the password policy
CORS_ORIGINS	Allowed web origins	* refused
PUBLIC_BASE_URL	Public API address for links	localhost refused
UPLOAD_DIR	File store path	
SMTP_URL	Mail account	
SEED_SAMPLE_DATA	Demo data on or off (off by default in production)	
SCHEDULER_ENABLED	Runs scheduled jobs on this instance	
<prefix>_SECRET_KEY, <prefix>_WEBHOOK_SECRET, <prefix>_MERCHANT_ID, <prefix>_PASSWORD	Payment gateway credentials, read only from the environment	A gateway cannot go live without them

If a rule fails, the API logs Refusing to start in production: ... with the reason and exits. Secrets are never stored in the repository; deploy/backend.env.example holds placeholders only.

7.4 Database migrations on start

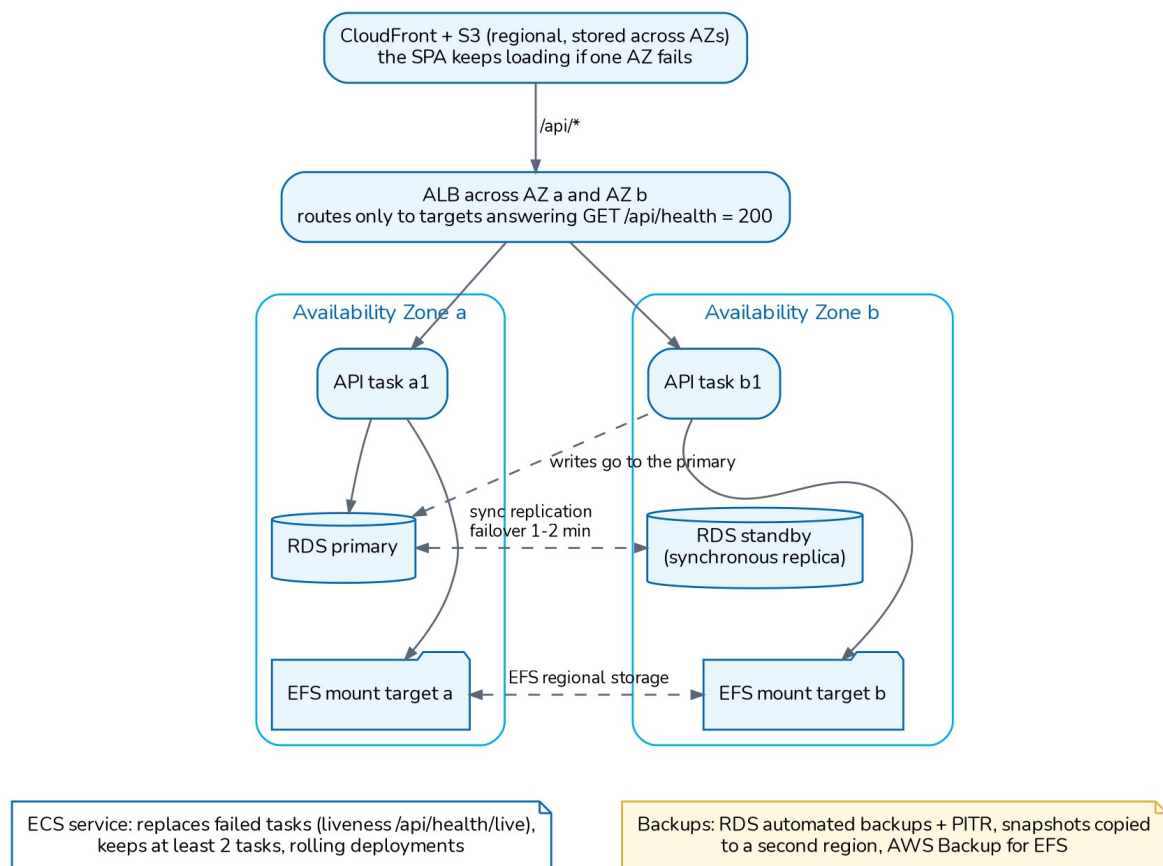
1. The server checks the production configuration.
2. It reports not ready (/api/health answers 503) and takes a PostgreSQL advisory lock. Other instances starting at the same time wait (up to 10 minutes).
3. It applies each pending migration in its own transaction and records it in schema_migrations, then runs the idempotent seed (roles, permissions, settings, chart of accounts, product templates, jobs, masters, administrator).
4. It starts HTTP and the scheduler and reports ready.

Migrations only add to the schema (project convention), so the previous release normally runs on the newer schema. A failed migration stops the start-up; the release is not rolled back automatically. **Recommended:** test every migration on a UAT copy and keep a pre-release snapshot.

8. High availability, backup and recovery

8.1 Resilience properties of the application

Property	Effect
Stateless API	Any instance serves any request; no sticky sessions
Readiness and liveness	/api/health checks the database (2,000 ms timeout) and pending migrations; /api/health/live checks the process only
Graceful shutdown	On SIGTERM: not ready, scheduler stopped, requests drained, forced exit after 25 s
Jobs on several instances	Advisory lock per run; a job runs once even with several schedulers
One transaction per business operation	A failure rolls back the whole operation including the document number
E-mail outbox	SMTP outages delay e-mail without failing transactions



High-availability topology (architecture document 07, recommended)

Recommended production topology: at least two API instances in two availability zones, rolling deployment with 100% minimum healthy, managed PostgreSQL with a synchronous standby, a shared file store across zones, container stop timeout 30 s and load balancer deregistration delay 30 s.

Single point of failure	Mitigation
Database	Standby with automatic failover; PITR
File store	Shared, zone-redundant store with daily backup

Single point of failure	Mitigation
Single API instance	Two or more instances
Secrets	Versioned secret store and sealed escrow copy of DATA_ENCRYPTION_KEY and JWT_SECRET
Region or site	Cross-region or off-site copies and a documented recovery

Known limitation: Rate limits are counted per API process, so with N instances the effective limit is N times the setting. **Recommended:** enforce rate limits at the WAF as well.

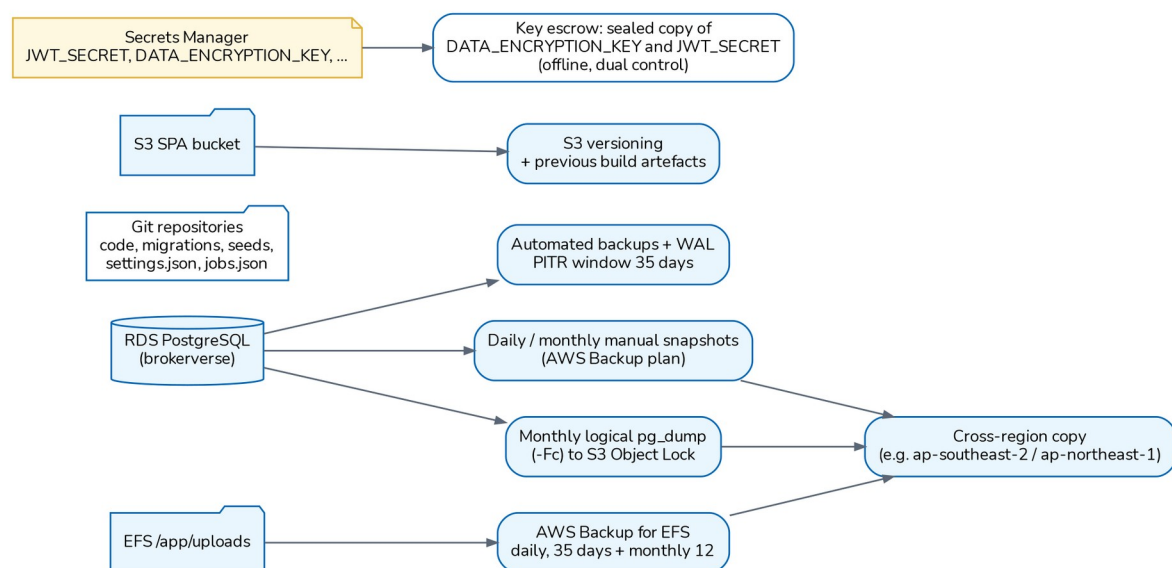
8.2 Recovery objectives

Recommended: The objectives below are the proposal of architecture document 08. Nothing in the code sets an RTO or RPO; the broker confirms them.

Scenario	RPO	RTO
API instance failure	0	under 5 minutes
Availability zone failure	0	under 15 minutes
Database instance failure (with standby)	0	under 15 minutes
Logical corruption or wrong mass change	15 minutes or less (5 achievable with PITR)	4 hours or less
Loss of uploaded files	24 hours or less	4 hours or less
Regional outage	24 hours or less	24 hours or less (warm standby: 8 hours)
Front end deleted	0	under 1 hour
Loss of secrets	not applicable	under 1 hour

Service tiers: Tier 1 core transactions in PostgreSQL; Tier 2 documents in the file store; Tier 3 derived data (notifications, outbox, generated reports); Tier 4 code and configuration as code, rebuilt from Git.

8.3 Backup



Backup and recovery data flows (architecture document 09, recommended)

deploy/README.md requires daily automated database snapshots kept at least 30 days, a backup of the upload volume and of DATA_ENCRYPTION_KEY, and one restore test before go-live. The code contains no backup automation. Document 09 recommends:

Backup	Retention (recommended)
Automated database backups with PITR, window 02:00 to 03:00 Manila time	35 days
Daily and monthly snapshots; monthly copy to a second region	Daily 35 days, monthly 12 months, yearly 10 years for year-end records
Snapshot before each release with migrations, before go-live imports and before each year-end close	At least 30 days
Monthly logical dump (pg_dump -Fc) to write-once storage	12 months
File store daily, weekly copy to the second region	Daily 35 days, monthly 12 months
Front-end bucket versioning	90 days
Container images	Last 10 releases

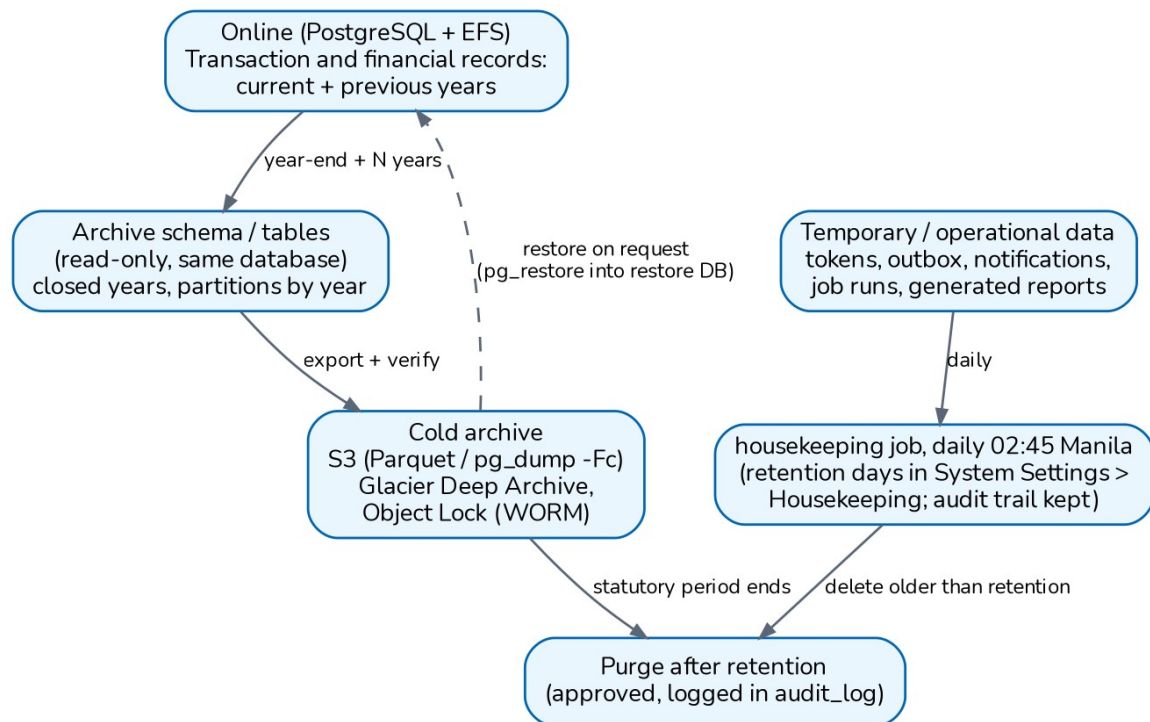
Restore procedures for PITR, snapshots, dumps, files, the front end, secrets and release rollback are in document 09. **Recommended** restore tests: PITR quarterly and before go-live; dump restore twice a year; file restore quarterly; secret escrow check yearly; full DR exercise yearly as a tabletop and every two years in full.

8.4 Archival and housekeeping

The daily housekeeping job (02:45 Manila time) deletes operational rows past the periods set in System Settings > Housekeeping, in batches of 5,000.

Data	Default retention
job_runs	90 days
email_outbox sent / failed	180 / 730 days
login_history	365 days
refresh_tokens expired or revoked	30 days
password_resets	7 days
Read notifications	180 days
Completed job_queue items	30 days
Generated reports (reports.retention_days)	90 days
audit_log	Kept (0); never purged below 2,557 days (seven years) if a period is set

Business and financial records are kept; financial documents are cancelled or reversed, not deleted. The dormant-users job (01:45 daily) deactivates accounts without a sign-in for access.dormant_days (90 days).

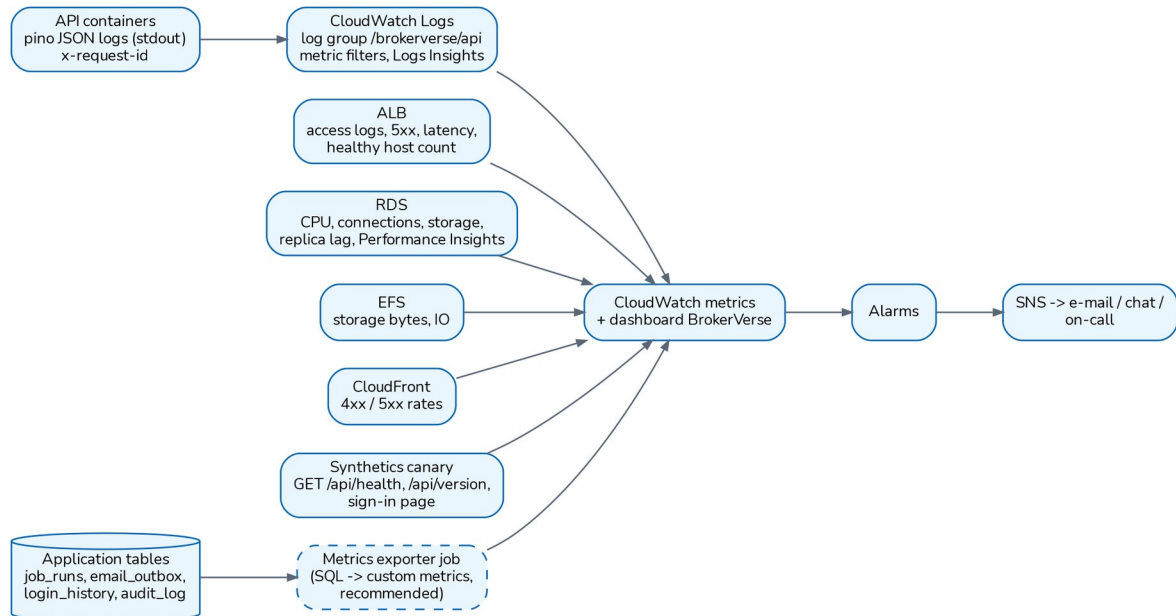


Data lifecycle: online, archive, purge (architecture document 10, recommended)

Recommended: Document 10 proposes keeping financial and policy records 10 years after the financial year end or policy expiry, unconverted leads and quotations 3 years then anonymised or deleted, and an annual export of closed years to write-once archive storage. The application anonymises the personal data of a client or prospect on request (Master > Data Privacy > Data Subject Requests, Anonymise, with a dry run first); it refuses while the records must still be kept (privacy.retention_years, 10 years after the last policy expiry). There is no archive function: the annual export of closed years is a database procedure run by the hosting team under change control.

8.5 Monitoring and alerting

The application produces JSON logs on standard output (pino) with a request id on every line and response, redaction of authorization headers, passwords, tokens and signatures, the health and version endpoints, and operational tables (job_runs, email_outbox, login_history, audit_log, generated_reports). The deployment package contains no monitoring configuration.



Monitoring and alerting (architecture document 11, recommended)

Alarm (recommended)	Threshold	Severity
Synthetic health check	2 consecutive failures, check every 5 minutes	P1
No healthy API instance	1 minute	P1
HTTP 5xx	over 2% for 5 minutes	P1
p95 response time	over 2 s for 10 minutes	P2
API CPU / memory	over 80% for 15 min / over 85% for 10 min	P2
Database CPU, free storage, connections	over 80% / under 20% / over 80% of maximum	P2
Scheduled job failed or not run	any failure of a daily job; over 26 hours since success	P2
Failed sign-ins	over 50 in 5 minutes	P2
Refresh-token reuse	any	P3
Backup failure or PITR older than 15 minutes	any	P2
Certificate expiry	under 30 days	P3

Severity P1 is a 24 x 7 response, P2 the same business day and P3 the next business day, to be aligned with the support agreement in docs/onboarding/SUPPORT_AND_ESCALATION.md.

9. Security architecture

9.1 Identity and authentication

Control	Implementation
Accounts	Named users only; no shared or agent sign-in. Users are created in Master > Generals > User Management or from a CSV with scripts/provision-users.js; each must change the initial password at first sign-in.
Password storage	bcrypt (cost 10); reset codes stored only as keyed HMAC hashes
Password policy	Minimum 8 characters, upper and lower case, digit and symbol, history of 5, maximum age 90 days (security.password_*)
Lockout	Account locked after 5 wrong passwords (limits.max_login_attempts); 10 attempts per 5 minutes per IP and per username (security.login_rate_limit)
Two-factor	TOTP (RFC 6238); secrets encrypted with AES-256-GCM; compulsory for the roles in security.require_2fa_roles (empty by default)
Sessions	Access token 30 minutes, refresh token 30 days rotated on each use; reuse of a rotated token revokes the token family and is logged
Revocation	Password change or reset, deactivation, role or permission change raise the token version and end sessions at once; Master > Generals > User Management can end every session of a user
Idle sign-out	30 minutes (limits.session_idle_minutes) with a one-minute warning
Sign-in history	Every attempt in login_history with result, reason, IP and user agent

Recommended: Before go-live set security.require_2fa_roles to at least system-admin, accounting and accounting-manager, as the deployment checklist suggests.

Known limitation: Tokens are kept in the browser's local storage (accepted for go-live in deploy/README.md section 9). A change to an httpOnly refresh cookie is planned. Two-factor enrolment shows the key and a link, not a QR code.

9.2 Authorisation, maker-checker and segregation of duties

Permissions are read:<module>, write:<module> and approve:<area> codes, checked by the server on every route (requirePermission, requireRole); the route registry adds the sign-in check to every route that is not explicitly public. The front end also hides menus by role (deny by default).

Role (seeded)	Main access
System Administrator (system-admin)	Every permission; configuration, users and roles
Sales & Marketing (sales)	Leads, clients, quotations, policies, endorsements, renewals, reports
Processing Team (processing)	Broker slips and placement, policies, endorsements, renewals, reinsurance, products
Operations (operations)	Leads, clients, quotations, policies, endorsements, renewals
Claims (claims)	Claims; read clients and policies
Accounting (accounting)	Receipts, collections, disbursements, commission, remittance, incentive, journals, period end, bank reconciliation
Accounting Manager (accounting-manager)	Inherits Accounting; approves period close, bank and insurer reconciliations, credit control, posting rule changes

Control	Implementation
Administrator protection	Only a System Administrator can grant that role or change an administrator; nobody can change their own roles or status

Control	Implementation
Record scope	Roles listed in <code>security.scoped_roles</code> see only their own records and clients (empty by default)
Maker-checker	A different user approves journal vouchers, payment vouchers and cheques, commission payouts, remittances and settlements, debit notes, claim settlements, incentive calculations, treaties, petty cash, period closes, reconciliations, quotation and renewal approvals, authority limits and posting rule changes (<code>finance.maker_checker_enabled</code>)
Authority matrix	Approval limits per transaction type and role or user (Master > Generals > User Management > Authority Matrix); a new limit applies only after another administrator approves it; default limits in PHP are seeded and replaced by the board-approved signing authority
Delegation of authority	Time-bound delegation of approval authority (leave, travel), revocable (Master > Generals > User Management > Delegations)
Segregation of duties	Rules on pairs of roles, checked when roles are assigned (<code>access.sod_enforced</code>). Seeded: placement and payment, placement and payment approval, claims and payment (block); sales and collection, sales and claims (warn)
Access reviews	Recertification campaigns over every active user; "revoke" deactivates the account and signs it out (Master > Generals > User Management > Access Reviews)
Access reports	User Access Matrix and Role Permissions, downloadable as XLSX or CSV
Period control	Postings refused in closed or locked periods; soft-closed periods only with <code>approve:period-end</code>

9.3 Application security controls (OWASP Top 10, 2021)

OWASP risk	Controls in BrokerVerse
A01 Broken access control	Authentication by default on every route; permission check per route; record scope; maker-checker; signed, expiring file links (30 minutes)
A02 Cryptographic failures	TLS in transit; bcrypt passwords; AES-256-GCM for TOTP secrets; HMAC-SHA256 for links, reset codes and webhooks; JWT limited to HS256
A03 Injection	Parameterised SQL only, dynamic identifiers from white lists; zod validation; CSV formula-injection guarding
A04 Insecure design	Maker-checker, authority limits, SoD, one transaction per operation, idempotent payment webhooks with amount check
A05 Security misconfiguration	Production start refuses unsafe secrets, <code>CORS_ORIGINS=*</code> and localhost URLs; helmet headers; x-powered-by off; sample data off in production
A06 Vulnerable components	Lock files with <code>npm ci</code> ; npm audit at review (see vulnerability management)
A07 Identification and authentication failures	Password policy, lockout, rate limits, TOTP, refresh-token rotation with reuse detection, idle sign-out
A08 Software and data integrity failures	Webhook signatures verified on the raw body; uploads checked by file signature; lock files; reviewed pull requests (recommended)
A09 Security logging and monitoring failures	Audit log, sign-in history, request ids, redacted JSON logs; alarms recommended in the monitoring chapter
A10 Server-side request forgery	The API calls only fixed provider addresses

OWASP risk	Controls in BrokerVerse
	(SMTP, PayMongo, Dragonpay); no user-supplied URL is fetched

Further controls: uploads limited by type (`uploads.allowed_types`, checked by file signature), size (10 MB, 10 files) and decompressed size (50 MB); stored files served with `nosniff` and a sandbox content security policy, and anything other than images and PDF as a download; 5xx responses return a generic message with the request id.

9.4 Encryption

Data	In transit	At rest
Browser to edge	HTTPS (CDN, load balancer or reverse proxy)	
Edge to API	HTTPS or private network	
API to database	TLS (<code>sslmode=require</code>)	Storage encryption of the managed database (KMS or provider key), recommended on every option
API to SMTP	STARTTLS on port 587	
File store		Storage encryption of EFS, Azure Files or the partner's storage (recommended)
Backups		Encrypted snapshots and write-once storage (recommended)
TOTP secrets		AES-256-GCM, key derived from <code>DATA_ENCRYPTION_KEY</code>
Password reset codes		HMAC-SHA256 keyed from <code>DATA_ENCRYPTION_KEY</code>
Sandbox gateway webhook secret		Derived from <code>DATA_ENCRYPTION_KEY</code> when no secret is set
Passwords		bcrypt hash

`DATA_ENCRYPTION_KEY` is not used to encrypt client records, ID numbers or uploaded documents; these rely on storage encryption of the database, file store and backups. Losing the key makes two-factor secrets unreadable (administrators reset two-factor and users enrol again); no business data is lost.

Recommended: Keep `DATA_ENCRYPTION_KEY` and `JWT_SECRET` in the secret store with versioning and an escrow copy under dual control. Turn on storage encryption for the database, file store and backups in every option; on premises use encrypted volumes (LUKS or the storage array's encryption).

9.5 Audit trail and logging

Record	Content
<code>audit_log</code>	User, entity, record id, action, before and after JSON, IP and time, written by every mutating handler; kept by default
Field-level and status history	<code>claim_field_changes</code> , <code>claim_history</code> , period status, bank reconciliation history, remittance approvals
<code>login_history</code>	Sign-in attempts, two-factor results, token reuse
<code>payment_events</code>	Every gateway notification with signature result
Application log	JSON on standard output with request id; secrets redacted

Master > Audit Trail filters by entity, record, user and date. **Recommended** reviews (document 11): monthly access reviews of administrators and new or dormant users; weekly review of changes to settings, posting rules, number series, tax codes and the Company master; monthly review of maker-checker exceptions; weekly review of security events. Ship application logs to central storage kept at least one year.

9.6 Vulnerability management

Item	Status
Backend dependencies	npm audit reported no advisories on 29 September 2026 (document 04)
Front-end dependencies	71 advisories at the review, almost all build-time dependencies of react-scripts 5.0.1; Create React App is no longer maintained. Move to Vite is the planned remedy.
Unused packages	js-cookie, react-pro-sidebar, web-vitals to be removed
Base images	node:22-alpine, nginx:1.27-alpine, postgres:16-alpine

Recommended: Run npm audit and an image scan (for example Trivy or the registry's scanner) in CI on every pull request; enable GitHub Dependabot alerts and secret scanning; patch critical findings within 14 days and high within 30 days; patch the operating system, PostgreSQL and nginx monthly.

9.7 Penetration testing approach

Recommended: No penetration test report is part of the repository. Commission an independent test before go-live and yearly, and after major changes.

Step	Content
Scope	Web application and API on UAT with production configuration, public paths (approval, payment checkout, webhooks), the hosting perimeter
Method	OWASP Web Security Testing Guide and OWASP API Security Top 10; authenticated tests with one user per role
Focus	Access control between roles and scoped users, maker-checker bypass, file link signatures, upload handling, token handling, rate limits, webhook forgery
Rules	Written authorisation by the broker and the hosting provider; test data only; no production personal data
Follow-up	Findings rated, fixed and retested; summary kept as evidence for the NPC and auditors

9.8 Secure development lifecycle

Practice	Current state
Code review	Pull requests run ci.yml (lint and 65 backend test files against PostgreSQL 16; front-end tests)
Tests	Backend integration tests with vitest and supertest on a real database; front-end tests with Jest
Lint rules	no-console outside scripts, no-unused-vars and eeqeq as errors
Secure defaults	Production start-up checks; sample data off in production; routes authenticated unless declared public
API inventory	npm run export:api writes OpenAPI, Postman and Excel lists of every route and its permission
Secrets	Never in the repository; placeholders in deploy/*.env.example

Recommended: Branch protection with required reviews and CI; tests restored in the deployment pipeline; dependency and image scanning; a short threat review for changes to authentication, payments and personal data.

10. Data privacy

10.1 Legal framework

Instrument	Relevance
Data Privacy Act of 2012 (Republic Act 10173)	Principles of transparency, legitimate purpose and proportionality; lawful criteria for processing; rights of data subjects; security of personal data; accountability for transfers
Implementing Rules and Regulations of the DPA (2016)	Detailed duties of controllers and processors, organisational, physical and technical security measures, outsourcing and subcontracting
NPC Circular 16-03, Personal Data Breach Management	Breach response team, notification to the NPC and to affected data subjects within 72 hours of knowledge, breach records and annual report
NPC Circular 2022-04	Registration of personal data processing systems and of the DPO with the NPC
Other NPC circulars and advisories	Security of personal data, consent, privacy impact assessment and administrative fines; the DPO confirms the current issuances

10.2 Roles under the DPA

Party	Role	Main duties
The broker	Personal information controller	Decides the purposes and means; appoints the DPO; registers with the NPC where required; answers data subjects; notifies breaches
iorta TechNXT	Personal information processor (when it hosts, supports or accesses production data)	Processes only on the broker's documented instructions; confidentiality; security; assists with rights requests and breaches
Cloud provider or hosting partner	Sub-processor	Infrastructure security under its contract
Insurers, reinsurers, payment gateways	Separate controllers or processors for their own processing	Covered by the broker's agreements with them

Recommended: Sign a Data Processing Agreement between the broker and iorta TechNXT covering the subject matter and duration, the instructions, confidentiality, security measures (this document), sub-processors (the hosting provider), assistance with rights requests and breaches (notice to the broker without undue delay so the 72-hour deadline can be met), return or deletion of data at the end of the contract, and audit rights.

10.3 Personal data held in BrokerVerse

Data subject	Data	Where
Leads and prospects	Name, company, e-mail, phone, birth date, gender, address	Leads
Clients (insured persons and companies)	Name, e-mail, phone, TIN, birth date, gender, address, credit limit	clients
Insured risks	Vehicle plate, chassis and engine numbers; property locations	Quotations, policies
KYC	Government ID type and number	Policies, uploaded

Data subject	Data	Where
	(PhilSys ID, UMID, passport, driver's licence and others in policy.kyc_id_types), ID image	documents
Claimants and payees	Loss details, description, settlement amounts, payee and bank details, claim documents and photos	claims, claim cash, documents
Payments	Payment references, amounts, gateway notifications (no card numbers: card entry happens on the gateway's page)	policy_payments, payment_events
Referrers and sub-agents	Name, bank account, withholding tax data	Referrer master
Staff users	Name, username, e-mail, roles, sign-in history with IP and browser	users, login_history

Government-issued identifiers and some claim information (for example injury details in a personal accident claim) are sensitive personal information under the DPA and need the stricter protection the DPA requires. The system holds no structured beneficiary register for policies; named beneficiaries, where captured, sit in policy details or documents.

10.4 Lawful basis and consent

Purpose	Usual basis under the DPA (to be confirmed by the DPO)
Quotation, placement, policy issue, endorsements, claims, renewals	Necessary for a contract with the data subject or steps at their request
Accounting, tax records, BIR forms, IC examination	Compliance with a legal obligation
KYC identification	Legal obligation and the insurers' underwriting requirements
Marketing to leads, win-back campaigns	Consent, or legitimate interest where the DPO concludes it applies

Consent is recorded per client or prospect and per purpose (processing, that is the privacy notice acknowledged; marketing; sharing with insurers and reinsurers) on the Data privacy tab of the client and on the prospect view, with the channel, the evidence and the notice version in force (privacy.notice_version). A withdrawal is stamped on the consent it ends; records are never deleted. Master > Data Privacy > Consent Register lists every consent across clients and prospects.

Recommended: the broker keeps the wording of its privacy notice in its own forms and proposal documents, raises privacy.notice_version when the notice changes, and checks the marketing consent before a campaign.

10.5 Data subject rights

Right (DPA section 16 and following)	How it is handled with BrokerVerse
To be informed	Broker's privacy notice; acknowledgement recorded as the processing consent with the notice version
Access	Export personal data (JSON or Excel) from the data subject request; client view with its policies; documents in the document viewers
Rectification	Edit the client or lead (Operations screens); every change is in the audit log

Right (DPA section 16 and following)	How it is handled with BrokerVerse
Object and withdraw consent	Withdraw the consent on the client or prospect; the request is logged with type Objection or Withdraw consent
Erasure or blocking	Anonymise from the data subject request (dry run first); refused while policy, claim and accounting records must be kept (privacy.retention_years). Leads can also be deleted (soft delete). Amounts, numbers and dates stay for the books.
Data portability	Export personal data in JSON or Excel
Damages and complaints	Broker's DPO and complaints procedure

Master > Data Privacy > Data Subject Requests is the register of requests: number from the DSR series, requester, type (Access, Rectification, Erasure or blocking, Objection, Data portability, Withdraw consent), date received and a due date `privacy.request_due_days` (15 calendar days) later, status, assignee, outcome and the exports and anonymisation done for it. The `privacy-requests-due` job (07:00 daily, delivered switched off) notifies the holders of `read:privacy` of open requests past their due date.

Recommended: the DPO confirms the response period and switches on the `privacy-requests-due` job at go-live (Master > Schedules).

10.6 Retention and disposal

Retention follows the recommendation of architecture document 10 until the compliance officer confirms it: financial and policy records 10 years, audit trail 10 years, unconverted leads and quotations 3 years then anonymised or deleted, security logs 2 years. Housekeeping already removes temporary data automatically (see the archival chapter). Disposal of media follows the hosting provider's certified destruction; on premises, use secure wipe or physical destruction with a certificate.

10.7 Security of personal data

The organisational, physical and technical measures required by the DPA and its IRR map to this document as follows.

Measure	Where addressed
Organisational: DPO, policies, training, contracts with processors	Broker; Data Processing Agreement (recommended)
Physical: data centre controls	Cloud provider or partner certifications
Technical: access control, encryption, audit, monitoring, backup, vulnerability management	Security architecture and resilience chapters

10.8 Cross-border transfer

When BrokerVerse is hosted in Singapore (options A and B), personal data is stored and processed outside the Philippines. The DPA allows this; the broker remains responsible and must use contractual or other reasonable means to give comparable protection. **Recommended:** record the transfer in the broker's privacy notice and processing records, rely on the provider's data processing terms and certifications, keep encryption keys under the broker's account, and document the decision in a privacy impact assessment. Option C keeps the data in the Philippines.

10.9 Breach management

Step	Practice with BrokerVerse
Detect	Alarms on failed sign-ins, token reuse, 5xx spikes; audit and sign-in history; WAF logs
Contain	End user sessions (Master > Generals > User Management), deactivate accounts, rotate JWT_SECRET to sign everyone out, block addresses at the WAF
Assess	Use the audit log, login_history and request ids to find which records were affected
Notify	The broker's DPO notifies the NPC and affected data subjects within 72 hours of knowledge where NPC Circular 16-03 requires it; iorta TechNXT informs the broker without undue delay
Record	Incident and breach records kept for the annual report to the NPC

11. Insurance Commission and records

The Amended Insurance Code (Republic Act 10607) and the circulars of the Insurance Commission require licensed brokers to keep complete and accurate books and records of the business they transact, to keep premiums received for insurers apart from their own funds and remit them on time, and to make their records available for examination by the Commission. The IC also expects regulated entities to manage information technology and cyber risk. BrokerVerse supports these expectations as follows; the compliance officer confirms the applicable circulars.

Expectation	Support in BrokerVerse
Complete records of policies placed, premiums, commissions	Policy, billing, receipt, remittance and commission registers with journals from posting rules
Premiums held for insurers and remitted	Premium trust bank account in the masters; remittance with multi-level approval; remittance ageing on the insurer's terms
Records available for examination	Reports in XLSX, CSV and PDF; audit trail; retention as recommended above
Reconciliation with insurers	Insurer statement reconciliation with approval and adjustment journals
Control over changes and approvals	Maker-checker, authority matrix, SoD, period locks
IT resilience and security	Backup, recovery objectives, monitoring and security controls in this document

12. Compliance checklist

Control	How BrokerVerse meets it	Customer responsibility
Named user accounts	Users created per person; initial password changed at first sign-in	Request, approve and remove access through a joiner-mover-leaver process
Strong passwords and lockout	Policy of 8 characters with four classes, 5-password history, 90-day age, lockout after 5 failures	Review the settings before go-live
Two-factor authentication	TOTP built in, compulsory per role	Set security.require_2fa_roles; reset two-factor when a phone is lost
Least privilege	Seven roles with read and write permissions; deny by default	Assign roles by job; keep few System Administrators
Segregation of duties	SoD rules checked on role assignment; maker-checker	Approve the rule set; act on conflicts in the User Access Matrix
Approval limits	Authority matrix with approved limits and delegations	Replace default limits with the board-approved signing authority
Access recertification	Access Reviews; dormant accounts deactivated after 90 days	Run a review at least twice a year (recommended)
Audit trail	audit_log with before and after values, kept by default	Review sensitive changes; set the retention period
Encryption in transit	HTTPS; TLS to the database; STARTTLS to SMTP	Provide certificates; refuse plain HTTP
Encryption at rest	TOTP secrets encrypted; passwords hashed	Turn on storage encryption for database, files and backups
Secrets management	Production start refuses weak or missing secrets	Store secrets in a vault; escrow DATA_ENCRYPTION_KEY
Backup and restore	Transactional database; health checks after restore	Configure backups; test restores quarterly
High availability	Stateless API, single job execution, graceful shutdown	Run two instances and a database standby
Monitoring	Health and version endpoints, JSON logs, operational tables	Set up dashboards, alarms and on-call
Vulnerability management	Lock files; backend audit clean at review	Scan in CI; patch OS and middleware; yearly penetration test
Change management	Pull request CI; migrations on start under a lock	Branch protection; UAT sign-off; pre-release snapshot
DPO and registration	Not a system function	Appoint the DPO; register with the NPC where required
Privacy notice and consent	Consent per purpose with notice version; Consent Register	Issue the notice; record consent; act on withdrawals
Data subject rights	Data Subject Requests register with due dates; export; anonymisation	Log every request; respond within the agreed period
Retention and disposal	Housekeeping of temporary data; financial records kept; anonymisation after the retention period	Approve retention periods; run archival
Cross-border transfer	Hosting choice of Singapore or the Philippines	Document the transfer; sign the provider's data processing terms
Data processing agreement	iorta TechNXT acts on instructions	Sign the agreement with iorta TechNXT and the hosting partner
Breach notification	Logs and audit data to assess a breach	Breach response team; notify NPC and data subjects within 72 hours
Insurer and IC records	Registers, reconciliations, reports	Keep records available for IC examination
Payment card data	Card details entered on the gateway's page only	Use gateways that hold the card industry certification

13. Document Sign-Off

Serial Number	Description	Owner	Status
1	No open items at the time of issue	iorta TechNXT	

Sign Off Date	Signature	Approved By	Department & Project Role



Insurance technology for the Philippines



10+
COUNTRIES
Global Presence



8
GLOBAL OFFICES
Strategically Located



11+
MARKETS SERVED
Across Continents



100%
COMMITMENT
To Excellence



MALAYSIA



INDIA



PHILIPPINES



THAILAND



VIETNAM



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ETHIOPIA



CAMBODIA



HONGKONG



SINGAPORE



Scan More Details